

Atharva Mavale

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CAREER PROFILE

- AI/ML engineer and Master of Applied AI student at Deakin University, building production-grade systems across MLOps, full-stack applications, and robotics.
- Delivered an Azure ML SDK v2 predictive maintenance pipeline with custom feature engineering and deployment to a managed online endpoint for real-time inference.
- Built a PyTorch EfficientNet-B4–based skin cancer risk assessment platform to enable accessible, early-stage medical risk screening via the web.
- Developing Fit Track, a FastAPI/React fitness tracking system with Alexa (AWS Lambda) integration; extending platform with AI-driven nutrition parsing and habit insights.

TECHNICAL SKILLS

- **Languages:** Python (Advanced), SQL, Java, R, JavaScript
- **AI/ML:** PyTorch, OpenCV, LangChain, YOLO, Hugging Face, scikit-learn, NLTK
- **DevOps** Docker, Microsoft Azure
- **Database** MySQL, MongoDB
- **Development:** FastAPI, React, Vite, Streamlit, ROS 2, Git, Agile/SCRUM
- **Data & Cloud:** Tableau, Render, Vercel, Netlify, REST APIs, Obsidian

PROFESSIONAL EXPERIENCE

AI Developer (Capstone) | InnovaAite, Australia | Oct 2025–Present [[GitHub](#)]

- Tuned a YOLO + OCR pipeline for whiteboard detection by analyzing low-recall failure cases, adjusting confidence thresholds, and planning dataset expansion.
- Prototyped audio feedback so detections are usable for end users and begin integration tests instead of staying at research level. Hardened a React Native navigation app with safe-exit confirmation, consistent back/forward flows, and multiple TypeScript/runtime bug fixes informed by OSRM routing research.

Software Engineer | Deakin Competitive Robotics | Aug 2025–Present [[GitHub](#)]

- Competed in the **2026 Australian Rover Challenge** against 17 university teams and 390+ students from Australia, Poland, Kazakhstan, Indonesia, and India, representing Deakin University and achieving a 12th-place finish; strengthened teamwork, field problem-solving, and high-pressure debugging skills.
- Completed the rover control GUI by integrating ROSbridge for teleoperation and contributing to autonomous rover and arm navigation using ROS 2 and MoveIt path planning.

EDUCATION

Master of Applied Artificial Intelligence (Professional) | Deakin University, Melbourne | 2024–2026

Bachelor of Science in Information Technology | Pillai College, Mumbai | 2020–2023 | 8.38 CGPA

PROJECTS

Fit Track – Voice-First Fitness & Calorie Tracker | Alexa, AWS Lambda (Python), REST API, React | 2026 – Present

- Designed an end-to-end system where Alexa logs workouts/meals via Lambda into a Render-hosted backend, with a web dashboard for daily calories and history. [[Frontend](#) · [Backend](#)]
- Actively maintained open-source project with ~80 clones and ~40 unique cloners over 14 days, continuously improving data accuracy (time zone, source mapping) and UX.

DermaGuard AI – AI Skin Cancer Screening App | React, FastAPI, PyTorch EfficientNet-B4, Google Places API | Mar 2026

- Developed a full-stack skin cancer screening platform with AI-based image analysis, automated PDF reporting, and hospital discovery features. [[GitHub](#)]
- Reduced barriers to early detection by offering free preliminary mole-risk screening for users who may avoid costly clinic visits.

StudyMate AI – Agentic Socratic Tutor | FastAPI, Streamlit, GPT-4o-mini, LangChain | Feb 2026 [[GitHub](#)]

- Innovated repo-aware AI tutor with 3-level adaptive teaching serving 5+ users; deployed FastAPI backend on Render with auto-scaling.
- Submitted to Sophiie AI Agents Hackathon —among 200+ Australian developers for \$5K prize competition.

Mine Safety RAG Copilot | FastAPI, React, Python, RAG, YOLO | Jan 2026 [[GitHub](#)]

- Replaced keyword search with NLP Q&A system citing 10+ manuals; reduced query time by 75% with 1,000+ chunk RAG pipeline.
- Integrating YOLO PPE detection for real-time safety monitoring across 3 camera feeds.

CERTIFICATIONS

Python Programming | Techgen Institute | Sep–Dec 2023 | 60-hour intensive covering OOP, data structures; built 5 projects

After Effects | Frameboxx Vashi | Mar–May 2024 | 40-hour course in motion design and compositing